

Row actions: Open · Download · Ask source · Archive. With 300 sources the list virtualizes and folders appear as chips above the tabs.

ResearchHub

Electronics L...
Owner · 3 members

Search

Home

Workspaces

Workspace

Overview

Documents3

Sources12

Ask AI

Analyses17

Members

Grounded in 12 sources

Settings

Help

ANAdam Nowak
adam.nowak@...
...

Electronics Lab — Team 4 > Sources

AN

KW

MK

Upload source

Upload sources

Drag & drop files here

or

Browse files

Up to 50 MB each

PDF

DOCX

XLSX

CSV

TXT

4 FILES

lecture_05.pdf

2.4 MB

Uploading

Uploading... 1.6 of 2.4 MB

sample_data.csv

48 KB

Processing

Processing source... reading rows 28 of 42

notes.docx

220 KB

Ready

3 pages read · ready to cite

scan_notes.pdf

9.8 MB

Failed

Retry

We couldn't read this PDF — it looks like a scanned image with no text.

You can close this window. Processing continues and you'll get a notification.

Done

Source status — component variants

Same row, four states. The chip always carries an icon and a word; failures say what happened in one bounded sentence and offer a way forward.

calibration.xlsx

XLSX · 22 KB

Uploaded

Safely stored. Waiting for a free reader.

sample_data.csv

CSV · 48 KB

Processing

Reading rows and columns. Page icon shuffles while it works.

smith_2025.pdf

PDF · 31 pages

Ready

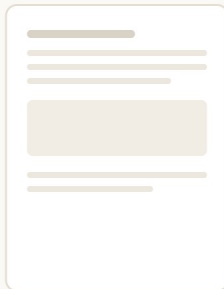
Searchable, citable by page. Appears in Ask scope.

scan_notes.pdf

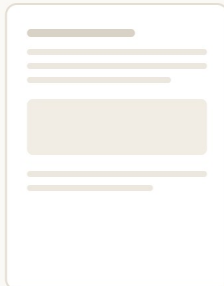
PDF · 9.8 MB

Failed

Scanned image, no text. Retry, replace, or run OCR later.



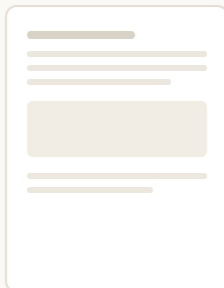
12



13



14



15



Title	Temperature dependence of silicon solar cell efficiency
Authors	Smith, J. · Okafor, L. · Tan, W.
Year	2025
Page	14 of 31
Section	3.2 Temperature dependence
Added	Kasia · Sep 29

Research actions

☆ Ask this source scope: 1 source

“ Cite p. 14

📄 Add to notes

🔍 Find related evidence 11 sources

🔍 Compare pick sources

Saved evidence · 2

“...conversion efficiency fell by 0.41 % per kelvin...”
p. 14 · cited in Laboratory Report [1]

“...open-circuit voltage dropped by 2.3 mV K⁻¹...”
p. 14 · not cited yet

3.2 Temperature dependence

Figure 6 summarises the efficiency of all five cells as a function of cell temperature. Between 25 °C and 75 °C, the efficiency of all cells decreased, with no evidence of a plateau.

Across the five cells, conversion efficiency fell by 0.41 % per kelvin over the 25–75 °C range (an average of 0.03 % K⁻¹). Most of this loss is attributable to the decrease in open-circuit voltage, which dropped by 2.3 mV K⁻¹, while the short-circuit current increased by only 0.05 % K⁻¹.

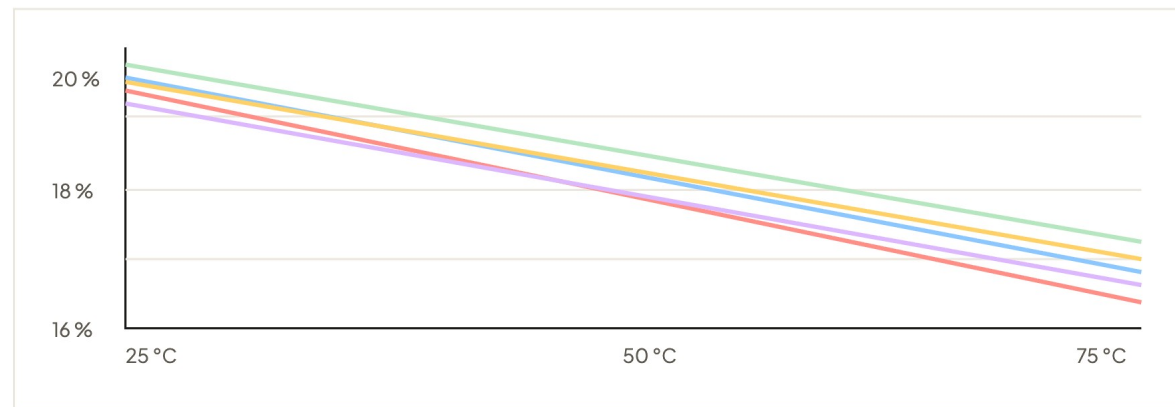
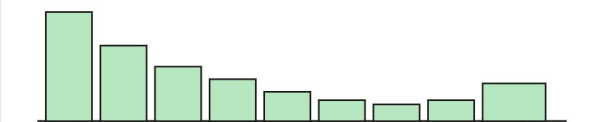


Fig. 6. Efficiency versus cell temperature for cells A–E under 1000 W m⁻² (AM1.5G).

These values are consistent with the temperature coefficients commonly reported for crystalline silicon technologies [18, 21]. The spread between cells (Table 4) is small compared with the effect of temperature itself, which suggests that the dominant mechanism is common to all devices.

For the practical design of cooled and uncooled modules this implies that a 40 K increase in operating temperature costs roughly one sixth of the nominal output.

supports claim [1] in Lab Report



⚠ Check before analysing

Used in

✦ Ask answer [3]
Rows 14–28 · measurement error

 **Laboratory Report**
Figure 1

 **Analyze**

- + Create analysis

#	123 frequency Hz	123 voltage V	123 current mA	abc note text · optional
8	1500	2.00	13.72	
9	1000	2.00	10.64	
10	700	2.00	8.05	
11	500	2.00	5.99	
12	350	2.00	4.30	
13	250	2.00	3.10	
14	200	2.00	2.49	first row below 200 Hz
15	170	2.00	2.12	
16	150	2.00	1.88	
17	130	2.00	1.63	
18	110	2.00	1.38	
19	100	2.00	1.25	
20	85	2.00	1.07	
21	70	2.00	0.88	
22	60	2.00	0.75	
23	50	2.00	0.63	
24	40	2.00	0.50	

Load more rows